ATLAS — Second Mathematical and Protocol Audit

Certified Self-Awareness as an Instrument of Automated Theorem Proving — Repaired v1.1

ATLAS-style independent static review

October 16, 2026

Final decision

MATHEMATICS: CONDITIONAL PASS. All five displayed propositions retain sound elementary proofs under their stated hypotheses. The two definitions and five research conjectures remain clearly identified. No gain in solver performance has been proved or measured.

MANUSCRIPT REPAIRS: PASS. All nine issues in the first ATLAS audit have been addressed at the level of manuscript wording, mathematical typing, causal interpretation, or future preregistration requirements. The repairs do not constitute successful execution of the required experiments.

EXECUTABLE EXPERIMENT: NOT YET AUTHORIZED OR REPRODUCIBLE. Prompts still contain genuine placeholders; no operational artifact set, checker configuration, fixed corpus, or new run trace is supplied. A proposed optional fifth arm is **not** authorized by this revision.

The original 13 major sections, 29 subsections, five propositions, five conjectures, five proofs, and four arms A–D are preserved in order. The corrected PDF builds as 17 pages (original 15). The corrected source has no unresolved LaTeX references or bibliography keys; the PDF opens and renders. No AMLD+0 run, solver substitution, verifier addition, target rewrite, or change to authorized experimental parameters occurred.

1. Reaudit of each mathematical proposition

P1 — No free reflective level: PASS (conditional, previously known). The countermodel witnesses the inner claim while excluding the outer claim from the epistemic domain. This disproves the unrestricted implication from awareness of a proposition to awareness of that awareness. The argument applies the previously cited *No-Free-+1* construction; it is not a novel independence theorem or a proof of machine consciousness.

P2 — Conservative certified reuse: PASS (conditional). Under the explicit premise that proof composition is sound and finite and the required hypotheses are identical or validly transported, replacing each appeal to the lemma with its certified derivation yields the target from the original assumptions. Nothing here proves proof-length reduction or net speedup.

P3 — Optional reflective actions preserve feasibility: PASS (conditional). An earlier certificate-producing path can be replayed if the old legal transitions, their effects, resource semantics, frozen target, and verifier are unchanged. This is an existence-of-path argument, not a guarantee about a scheduler or mandatory reflection overhead.

P4 — Gross value of optional information: PASS. For a finite action set and integrable utility, the posterior conditional maximum dominates the conditional expectation of any fixed action. Expectation and maximization over the fixed actions give the displayed inequality. The assumption that the controller can ignore information and does not pay for obtaining or using it is indispensable. Net utility may be negative.

P5 — Source-insensitive placebo limitation: PASS (narrow). A sound verifier cannot accept the same source-dependent assertion as true for two source instances when its truth values differ. This detects a class of source-insensitive claims; it does not establish a complete awareness test or useful control.

Proof tally: Five conditional valid arguments, zero supplied independent machine-checking certificates for the revised manuscript, zero observed causal effects. The five statements labeled Research Conjecture remain unproved.

2. Finding-by-finding closure

ATLAS-01 — HIGH — verifier typing: REPAIRED IN TEXT. Section 3 now distinguishes the fixed original object-level checker from proposed separately typed source, method, and receipt judgments. All auxiliary branches require prior authorization under the governing verifier. Source and receipt metadata must not become object-level BANK axioms by default. **Residual:** no authorized, executable auxiliary checking branch is included.

ATLAS-02 — HIGH — source and execution identity: REPAIRED. The model and Prompt P1 distinguish actual prompt bytes from their digest, executable build identity from a running process, and a run's manifest/ID from its actual append-only event trace. Acceptance records are conditional on an observed checker decision, not a claimed hash or self-report. **Residual:** the runtime manifest and actual trace remain placeholders until authorized execution.

ATLAS-03 — HIGH — causal interpretation: REPAIRED WITH A PROTOCOL LIMIT. Section 6 correctly describes A–D as reflective generation by telemetry-driven adaptation, not reflective generation by permission to read receipts. The difference-in-differences measures an interaction under declared assumptions, not the isolated value of receipt consumption. A receipt-blind fifth comparator E is described solely as an **unapproved option**, with explicit warnings that records in diverging runs need not be identical. **Residual:** isolating content-consumption causally requires separate user permission and a frozen intervention plan; A–D alone cannot establish it.

ATLAS-04 — HIGH — first-person claim: REPAIRED. Each approved experiment must retain its own exact first-person future sentence. Examples are explicitly noninterchangeable. No statement in the paper overrides a separately frozen first-person target.

ATLAS-05 — MODERATE — reflection depth: REPAIRED DEFINITION. Section 9 operationalizes accepted receipt depth as authenticated, well-typed, verifier-accepted, hash-linked sequential records and expressly separates this from controller rank, proof-theoretic reflection, and OSA levels. **Residual:** the actual test must freeze its precise checker implementation and costs.

ATLAS-06 — MODERATE — confounding: ADDRESSED AS PREREGISTRATION REQUIREMENTS. The manuscript now calls for source/knowledge access matching, the task population, sample size, seed pairing, accepted BANK baseline, fully charged costs, a single prespecified primary metric, uncertainty estimates, and timeout treatment. It warns against causal comparisons restricted to post-treatment successful runs. **Residual:** none of these parameters is instantiated or empirically checked.

ATLAS-07 — MODERATE — executable completeness: HONESTLY UNRESOLVED. The prompts remain templates with unresolved variables. There is no full experiment bundle with fixed task instances, validators, source manifests, runtime, and trace. The manuscript now makes this limitation conspicuous instead of describing placeholders as a ready experiment.

ATLAS-08 — MODERATE — falsifiability: ADDRESSED. Section 9 limits conjectural test setups to no-oracle, nontrivial, fully charged, equal-input-rights conditions and distinguishes existential conjectures from a task-family-specific falsifiable hypothesis with a nonzero threshold. **Residual:** the family, threshold, endpoint, and uncertainty standard must still be frozen before a trial. Negative findings on one sample do not logically refute a purely existential conjecture.

ATLAS-09 — LOW — originality: STILL OPEN, CORRECTLY QUALIFIED. The note makes no verified claim of global literary priority. The propositions include established logical and decision-theoretic observations. This second audit is not a literature novelty determination and does not present them as new discoveries.

3. New second-pass consistency checks

1. **One governing verifier.** Auxiliary validators are prospective, must be preauthorized under the same governing verifier, and do not implicitly expand the original theorem checker. An external reviewer alone does not promote claims into certified BANK.
2. **Typed storage.** Validated source/receipt events are metadata unless separately backed by an admitted mathematical proof. Observations and heuristics remain non-theorem data.

3. **Receipt depth.** The definition uses *well-typed* (not pairwise different type tags) records, so arbitrary finite chains are not excluded just because there are only three sample type categories. Counting starts from actual accepted events.
4. **Paired control.** The optional E design specifies matched generation procedures and a preregistered point of intervention; it does not falsely assume two diverging trajectories produce identical future receipts.
5. **Temporal run identity.** Prompt P1 can refer to an append-only trace prefix while the run is active. A completed execution trace is an artifact only after termination.
6. **No silent protocol modification.** All new experimental suggestions are explicitly prospective. The original arms, freeze boundaries, target, costs, agent roles, and user approval requirements remain protected.

4. Residual blocking conditions before ANY experiment

No prompt in the corrected paper constitutes authorization to start AMLD+0, FORGE-6, a chess engine, or any other prover. Before a valid experimental claim, the researcher must expressly approve the exact sentence/target and signed source manifests, the actual typed checker implementations and authorized branches, the task corpus and knowledge rights, the study arms (including a separate decision on optional E), the complete charged cost ledger, the seed and budget allocation, the registered analysis, and immutable logging. Every reported mathematical settlement must be accepted by the frozen actual object verifier. A receipt about a proof is not a replacement for the proof.

Even after these conditions are met, a positive experimental result would need external verification, reproducibility, and a meaningful comparator before being described as a verified performance gain. Novelty compared with external work requires a separate bibliographic audit.

5. Release checks and provenance

- Original ZIP source SHA-256: `8d16700d5de1ac411e2a01ff2dd0ec3a2dba4f8244a05d76c1abf91880a4e243`.
- Repaired v1.1 source SHA-256: `bc2fe2977cdc94f7118afb895a425795fd6ecbfd68df51045b77458f589f1807`.
- LaTeX build: successful with `latexmk`; no fatal errors, undefined cross-references, or undefined citations. Minor underfull line warnings only.
- PDF inspection: 17 readable letter-sized pages, not encrypted, searchable text; figures and tables rendered.
- Preservation: 13/13 original major sections in order; 29/29 original subsections in order; five proof statements/proofs and five conjectures remain present; arms A–D remain.

ATLAS final verdict: REPAIRED WORKING NOTE APPROVED FOR CONDITIONAL MATHEMATICAL CONSISTENCY AND CLEARLY LABELED PUBLICATION AS A PROPOSED RESEARCH PROGRAM. NOT AN EXECUTABLE OR VALIDATED PERFORMANCE EXPERIMENT. No independently verified improvement in ATP performance follows from this audit.